

Auditing Algorithmic Visibility on X: Evidence from Shadowban Detection

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Abstract

Shadowbans are a content moderation mechanism applied by social media platforms that reduces the reach of accounts or posts without explicit notification, making their effects difficult for users to detect or interpret. In this paper, we present a systematic audit of multiple forms of shadowbans on X, examining both account-level and post-level visibility restrictions using data from active users in 2025. Although shadowbans are rare overall, we find that they are structured rather than random, with different moderation logics governing search bans and reply bans. Search bans respond primarily to lifetime account signals, while reply bans are driven by recent activity, engagement, and content characteristics. These differentiated and opaque mechanisms illuminate why users experience visibility changes as confusing and unpredictable. Our findings highlight how limited feedback and fragmented enforcement signals complicate user sense-making, perceived legitimacy, and agency, underscoring the importance of transparency and interpretability in platform governance.

CCS Concepts

• **Human-centered computing** → **Collaborative and social computing**.

Keywords

Shadowban, Moderation, Algorithm Audit, Platform Governance, User Experience

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1 Introduction

Social media platforms function as sociotechnical infrastructures that mediate public discourse, political expression, and everyday participation. Users' ability to speak, be heard, and assess the reach of their contributions is shaped by algorithmic systems governing visibility, interaction, and feedback [6]. To manage public harms such as spam, harassment, and misinformation, platforms deploy a range of algorithmic moderation mechanisms [5].

One prominent example of such opacity is *shadowbanning*, a form of moderation in which the visibility of an account or its content is selectively reduced without notifying the affected user. Unlike account suspension or content removal, shadowbans preserve outward signs of participation while altering reach in ways that are difficult to detect or verify through user-facing signals. As a result, it remains challenging to assess when, how, and why shadowbans are applied.

Algorithmic visibility plays a central role in shaping perceived credibility and attention, even without explicit accuracy cues [10, 13, 17]. These challenges have become especially salient on X. Following changes in ownership and leadership, the platform has asserted that it no longer engages in covert or politically motivated moderation; however, perceptions of shadowbans persist [1, 4, 16]. Prior research shows that disputes over shadowbanning often become contests over credibility, with users attempting to make suppression legible in the absence of authoritative signals [2, 14]. Despite growing debate, systematic evidence about the contemporary deployment of shadowbans on X remains limited.

Although Twitter released portions of its recommendation algorithm source code in March 2023 [20], the disclosure does not clarify the parameters used for suppressing content. We, therefore offer a human-centered perspective in terms of how shadowbans are perceived, through a systematic audit of multiple types of shadowbans on X, enforced at the account-level and post-level, using data collected from active users in 2025. We find that shadowbans on X are rare overall, with low prevalence of search bans and reply bans. Moreover, search bans are associated primarily with long-term account characteristics (e.g., account age, follower count, verification), whereas reply bans are linked to recent activity patterns, engagement, and URL-sharing behavior. The findings empirically ground a contemporary understanding of the opaque moderation practices used in platform governance at X.

2 Related Work

2.1 User experiences of opaque moderation

A substantial body of research examines how users experience opaque forms of platform governance, particularly when moderation decisions are not explicitly communicated. Qualitative studies show that users often develop folk theories to explain unexplained changes in visibility, engagement, or account status [12]. More recent work focuses directly on shadowbanning and related visibility restrictions, documenting how users attempt to detect, theorize, and contest shadowbans through collective sensemaking, experimentation, and informal evidence sharing [3, 14]. This literature highlights the emotional, cognitive, and labor-intensive burden



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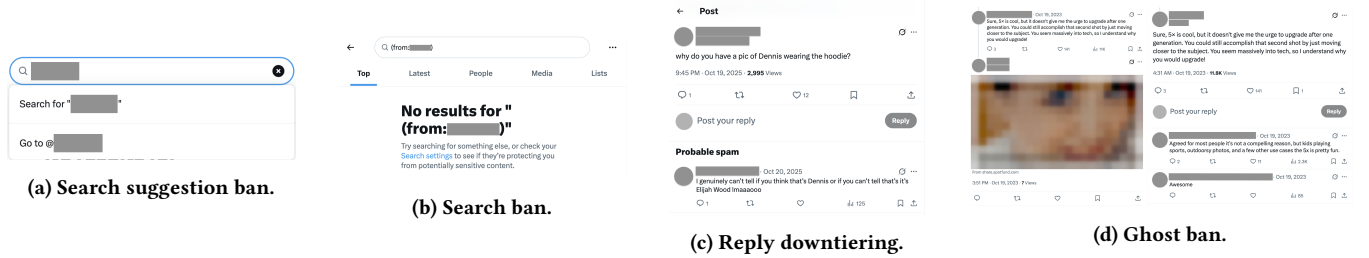


Figure 1: Illustrations of visibility restrictions on X. Account handles and profile pictures are blurred for privacy reasons.

placed on users—especially creators and marginalized communities—who must invest sustained effort to maintain visibility under uncertain platform rules. Related work further shows how creators adapt their practices in response to perceived algorithmic penalties, engaging in forms of invisible labor to mitigate or work around shadowbans [9].

Complementing empirical audits [7], this work emphasizes the role of user beliefs in platform economies: uncertainty about suppression can shape participation and welfare outcomes even when visibility restrictions are not directly observable. Together, these studies establish shadowbanning as a lived and contested phenomenon shaped by user interpretation. However, they primarily focus on experience and meaning-making rather than systematic measurement, leaving open questions about how visibility restrictions are applied at scale and how consistently they operate across users and content.

2.2 Measuring Shadowbans at Scale

Compared to the growing qualitative literature, fewer studies have attempted to measure shadowbans quantitatively, largely due to methodological and access constraints. Existing audits typically rely on indirect, behavior-based signals—such as search results, reply ordering, or engagement patterns—to infer visibility restrictions. For example, Le Merrer et al. [11] show that shadowbans on Twitter are associated with users’ interaction networks, while Jaidka et al. [8] demonstrate that bot-like behavior, offensiveness, partisan content, and engagement patterns predict different forms of visibility suppression.

These studies predate Elon Musk’s acquisition of Twitter and the subsequent reconfiguration of the platform (now X). Since then, X has articulated an enforcement philosophy emphasizing “freedom of speech, not reach” [18], while simultaneously reducing API access and limiting third-party measurement tools. As a result, prior findings may no longer reflect current moderation practices, and up-to-date empirical evidence on contemporary shadowban deployment remains scarce.

Beyond prevalence, visibility-based moderation has broader implications for information evaluation and trust. Research on online credibility shows that exposure and amplification play a central role in shaping perceived legitimacy of content and speakers [10, 17]. Even subtle changes in visibility can influence what users attend to, trust, and share, independent of content accuracy [13]. There is therefore a need to understand how shadowbans affect the informational signals audiences rely on to assess relevance and credibility.

At the same time, shadowbans should be treated as heterogeneous interventions. Platform enforcement increasingly differentiates between account-level visibility limits and post- or reply-level suppression, yet empirical work capturing these distinctions remains limited, in part due to data access constraints.

In sum, prior work places shadowbans at the intersection of technical enforcement, user interpretation, and information credibility, yet empirical evidence linking contemporary shadowban practices to observable visibility outcomes on X remains limited. Building on prior audit-based approaches, we examine both account-level and post-level shadowbans under current platform conditions.

3 Methods

3.1 Data Collection

Guided by X’s documented enforcement options [19], we focused on visibility-based moderation outcomes that are comparatively feasible to collect and interpret. Specifically, we examined three forms of shadowbans—search suggestion bans, search bans, and reply bans—which operate at different points in users’ discovery and interaction pathways.

Search suggestion bans. A search suggestion ban occurs when an account does not appear in the auto-complete suggestion prompt when its handle is queried. To measure this outcome, we queried each account handle from a newly created account with no prior activity, minimizing engagement-based personalization. Accounts that failed to appear in the suggestion prompt were labeled as search suggestion-banned.

Search bans. A search ban refers to the exclusion of a user’s posts from search results, including hashtag searches. We assessed search bans using advanced search queries of the form (from: handle) and verified whether recent posts were returned. Accounts whose posts did not appear in search results were labeled as search-banned.

Reply bans. Reply bans restrict the visibility of users’ replies within parent threads, either by making replies invisible to other users (commonly referred to as ghost bans) or by hiding them behind a “Show probable spam” interstitial (reply downtiering). Due to recent platform changes, reliably distinguishing between these two mechanisms is no longer technically feasible. We therefore treated them as a single category, *reply bans*. To measure reply bans, we collected up to five recent replies per user and checked whether each reply was visible within its parent thread. To reduce ambiguity in reply ordering, we restricted analysis to parent posts with fewer than 20 replies. If a user has fewer than five recent replies, all available replies are included.

3.1.1 Sampling. Public post and account metadata were collected using a combination of custom Python scripts applying the selenium package, and an independent third-party data access service that provided programmatic access to publicly available X content, in accordance with platform terms and rate limits. Visibility checks were conducted by initializing a new user session for each query to minimize the influence of prior activity or personalization.

Our initial analysis sought to replicate prior measurements using a set of users audited in 2022 and generously shared by prior work [8]. However, 11,512 of the original 23,814 accounts no longer existed. Among the remaining 12,302 accounts, only 2602 posted in 2025, and only 1289 had recent replies. We therefore collected a more recent sample of 12,351 users who posted geolocated (United States) content in early December 2025. Shadowban status was assessed during the week of December 15, 2025.

3.2 Feature Extraction and Operationalization

We extracted account- and post-level characteristics to examine whether they predicted shadowban outcomes. For account-level analysis, we considered user profile attributes and recent posting history; for post-level analysis, we additionally included characteristics of the parent post and parent author.

User profile features. For each account, we collected verification status, follower and following counts, number of favorited posts, total status updates, media count, and account age.

User posting histories. We collected each user’s posting history over the two weeks preceding the shadowban check. In this exploratory audit, we focused on a short observation window, similar to prior work [8, 15]; multi-wave data collection over longer time horizons is left to future work.

From these data, we computed posting frequency by type (retweets, quotes, replies, and original posts) and average engagement per non-retweet post (retweets, quotes, replies, likes, and bookmarks), serving as a proxy for popularity. We also characterized the credibility and political orientation of URLs shared by each user. Domains are mapped to Media Bias Fact Check (MBFC) credibility ratings and ideological labels, and per-user counts are computed for aggregated categories: *left*, *right*, *pro-science*, *High-cred*, and *Low-cred*.

Parent post and author features. For post-level reply-ban analysis, we collected engagement metrics for both the parent post and the reply, as well as observable attributes of the parent author.

Operationalization. Posting-history measures were computed over a 14-day window. Continuous count-based covariates were log transformed to reduce skewness, while MBFC-based URL counts were left untransformed due to sparsity.

3.3 Modeling

We estimated generalized linear models (GLMs) with a binomial family and logit link to examine associations between extracted features and shadowban outcomes, using ridge regularization. Regularization strength was tuned via 10-fold cross-validation with AUC as the optimization criterion. Covariates with no variance were excluded, and only non-zero coefficients are reported.

For account-level outcomes, we estimated:

$$\text{Shadowban}_i \sim \text{Bernoulli}(p_i),$$

$$\text{logit}(p_i) = \alpha + \mathbf{x}_i^{\text{Profile}} \boldsymbol{\beta} + \mathbf{x}_i^{\text{History}} \boldsymbol{\gamma}. \quad (\text{I})$$

For post-level reply-ban outcomes, we estimated a logistic model with author fixed effects:

$$\text{ReplyBan}_{jt} \sim \text{Bernoulli}(q_{jt}),$$

$$\text{logit}(q_{jt}) = \alpha_{u(j)} + \mathbf{x}_{jt}^{\text{ParentPost}} \boldsymbol{\beta} + \mathbf{x}_{jt}^{\text{ReplyPost}} \boldsymbol{\gamma} + \mathbf{x}_{jt}^{\text{ParentAuthor}} \boldsymbol{\delta}. \quad (\text{II})$$

4 Results

4.1 Descriptive Statistics

Across sampled users, we observe no instances of search suggestion bans. In contrast, 111 users (0.90%) are search banned, and 348 users (0.28%) have at least one reply-banned post. Overlap between these categories is minimal: only 6 users experience both. At the post level, 499 of 34,414 posts are shadowbanned (1.45%).

Overall, 20.81% of users are verified. Users have a median of 581 followings and 474 followers, with medians of 8918 favorites and 6449 status updates. Accounts are relatively mature, with a median age of 3472 days (approximately 9.5 years).

During the observation window, users are more reply-oriented than original-post oriented (median original posts = 4, replies = 13, quotes = 0). Median engagement per post is low (1.19). URL sharing is sparse: over half of users share no URLs, and among those who do, more than half share none that can be matched to Media Bias Fact Check (MBFC) labels, limiting domain-level coverage.

Parent authors are substantially more prominent than reply authors: 39.24% are verified, with a median account age of 3061.94 days (approximately 8.4 years), 2921 followers, 841 followings, and 19,395 status updates. This suggests that reply-banned content often targets older, more established accounts.

4.2 Modeling Results

Figure 2 reports estimated coefficients between shadowban outcomes and account- and post-level covariates.

Search ban. Account age was negatively associated with search bans ($\beta = -0.795$, 95% CI $[-0.985, -0.605]$), while follower count ($\beta = 0.201$, 95% CI $[0.022, 0.381]$), total status updates ($\beta = 0.286$, 95% CI $[0.089, 0.483]$), and verification ($\beta = 0.776$, 95% CI $[0.318, 1.230]$) were positively associated with search bans.

Posting histories indicate that higher volumes of original posts ($\beta = -0.699$, 95% CI $[-1.280, -0.114]$) and replies ($\beta = -0.726$, 95% CI $[-1.030, -0.424]$), as well as higher average engagement ($\beta = -0.910$, 95% CI $[-1.310, -0.514]$), were negatively associated with reply bans. Sharing pro-science URLs was positively associated with search bans ($\beta = 2.228$, 95% CI $[0.261, 4.200]$); other URL-ideology categories were not significant.

Reply ban (account level). Older accounts were less likely to be reply banned ($\beta = -0.159$, 95% CI $[-0.273, -0.045]$), while higher status update counts increased risk ($\beta = 0.122$, 95% CI $[0.008, 0.236]$). Follower count ($\beta = -0.145$, 95% CI $[-0.252, -0.038]$) and media count ($\beta = -0.106$, 95% CI $[-0.177, -0.035]$) were negatively associated with reply bans. Among posting-history features, quoting ($\beta = 0.275$, 95% CI $[0.074, 0.476]$), average engagement ($\beta = 0.177$, 95%

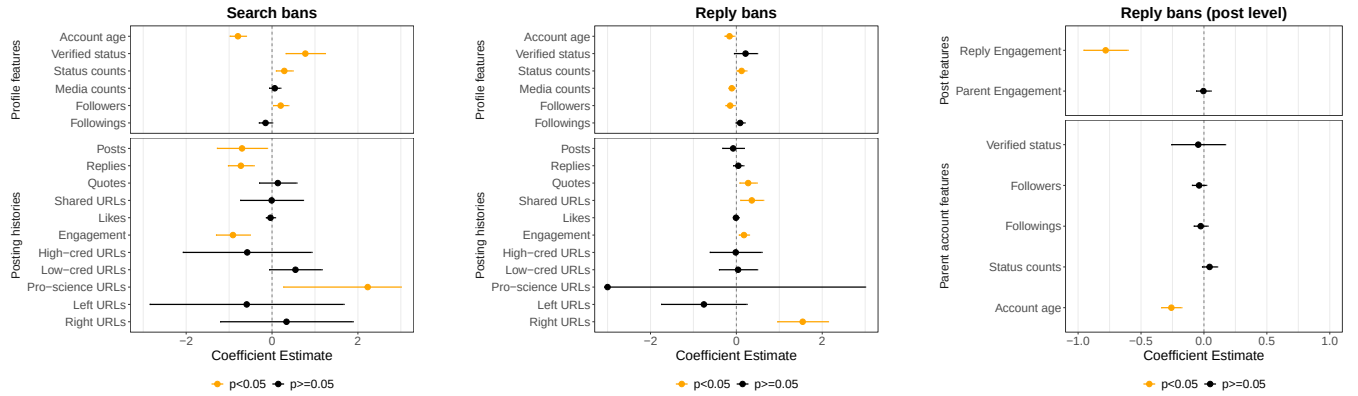


Figure 2: Regression results for search bans and reply bans. Markers denote coefficient estimates and error bars indicate 95% confidence intervals. At the account level, both search ban and reply ban can be predicted from different profile and posting features (yellow bars). At the post level, reply ban can be predicted by engagement and the age of the parent post.

CI [0.058, 0.296]), URL sharing ($\beta = 0.360$, 95% CI [0.092, 0.627]), and sharing right-leaning URLs ($\beta = 1.545$, 95% CI [0.954, 2.140]) were positively associated with account-level reply bans.

Reply ban (post level). Higher reply engagement ($\beta = -0.781$, 95% CI [-0.957, -0.606]) and replying to older parent posts ($\beta = -0.259$, 95% CI [-0.339, -0.179]) were negatively associated with reply bans.

5 Discussion

In this paper, we analyzed shadowbans on X at both the account and post levels. Consistent with prior work, we find that shadowbans are rare overall (0.90% search bans, 1.45% post-level), and that older accounts face lower risk across ban types [8]. On the one hand, search bans respond to lifetime account signals: older age ($\beta = -0.795$), follower count ($\beta = 0.201$), posting volume ($\beta = 0.286$), verification ($\beta = 0.776$), pro-science URLs ($\beta = 2.228$). On the other hand, reply bans track recent activity: quoting ($\beta = 0.275$), URL sharing ($\beta = 0.360$), right-leaning URLs ($\beta = 1.545$). At the post level, higher engagement ($\beta = -0.781$) and older parent accounts ($\beta = -0.259$) protect against reply bans, suggesting contextual legitimacy signals. Our results extend prior work by showing verification now predicts search bans (post-subscription changes) and enforcement varies by content: pro-science vs. right-leaning domains. These patterns indicate search bans constrain systemic discoverability (account-level risk) while reply bans enable context-sensitive conversational gatekeeping—optimized for distinct moderation objectives, though intent remains unobservable. We anticipate that this dual structure allows X to throttle high-risk accounts' broad reach without disrupting their conversational participation. Over time, such layered moderation may fragment users' social networks, reducing civic engagement as users withdraw from platform discourse amid perceived unfairness and eroded trust.

Taken together, our findings explain shadowbans' opacity, complicating user attribution and adaptation as documented in qualitative work. Even when rare, selective application shapes credibility and fairness perceptions. More broadly, mechanism-specific

moderation may reorganize amplification, reply visibility, and interactional hierarchies in ways users struggle to detect, warranting study of network/discourse impacts.

Limitations and future work. Our approach has faced several limitations. First, shadowban statuses are inherently difficult to collect. Our measures operationalize collection by measuring visibility outcomes, such as account discoverability or reply visibility within threads. Not all the visibility restrictions can be gathered. For instance, X may remove rule-violating posts from the For You and following timelines, or put violating accounts in read-only modes. Yet, there is no reliable way to gather such data. We argue that perfect data collection is impossible without platform-wise support, and the current data collection is the best we can do as researchers. Second, X's enforcement policies can be dynamic. Therefore, our observations are time-bound with respect to the platform's visibility governance.

Finally, this study analyzes a single snapshot of users and posts. A natural next step is to extend data collection into multiple waves. By doing so, we can examine the persistence of shadowbans and whether prior shadowban history predicts subsequent enforcement.

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